

NETWORK TYPES

Complete Study Guide with Diagrams

Learning Simple English

Key Topics Covered:

PAN • LAN • WLAN • CAN • MAN • SAN • WAN

What Are Network Types?

Networks are groups of computers and devices connected together. Networks can be different sizes. Some networks are very small (for one person), some are medium (for an office), and some are very large (covering entire countries or the whole world). The size and area they cover decide what type of network they are.

We Will Learn About 7 Types of Networks:

1. **PAN** - Personal Area Network (smallest)
2. **LAN** - Local Area Network (office or home)
3. **WLAN** - Wireless Local Area Network (without cables)
4. **CAN** - Campus Area Network (university)
5. **MAN** - Metropolitan Area Network (city)
6. **SAN** - Storage Area Network (for data storage)
7. **WAN** - Wide Area Network (largest, like the Internet)

1. PAN (Personal Area Network)

What is PAN?

A PAN is a small network used by one person. It connects personal devices together so they can talk to each other and share information.

Simple Example: Your phone, tablet, and laptop all connected together.

Devices in a PAN:

- Smartphones
- Tablets
- Laptops
- Smartwatches
- Wireless headphones

How They Connect:

Wireless Ways: Bluetooth, Infrared, NFC (Near Field Communication)

Wired Way: USB cables

What Do We Send?

- Music files
- Photos
- Calendar events
- Small documents

Size: Very small - Usually a few meters (10-30 feet)

Simple PAN Diagram:

■ Smartphone | Bluetooth/USB / | \ / | \ ■ ■ ■ Watch Laptop Headphones

2. LAN (Local Area Network)

What is LAN?

A LAN connects computers and devices in the same building or close area. All the devices are near each other (in the same office, school, or home).

Simple Example: All computers in your school connected together.

Devices in a LAN:

- Desktop computers
- Laptop computers
- Printers
- Servers
- Switches

How They Connect:

Most Common: Ethernet cables connected to a switch or router

Size: Usually covers one building or one area - up to 100 meters

3. WLAN (Wireless Local Area Network)

What is WLAN?

A WLAN is exactly like a LAN, but WITHOUT cables. It uses wireless connection instead.

Simple Example: WiFi in your home or school.

How It Works:

- A WiFi router sends wireless signals
- Your devices receive these signals
- No cables needed!

Devices in a WLAN:

- Laptops with WiFi
- Smartphones
- Tablets
- Wireless printers
- WiFi routers or Access Points

Key Difference From LAN: LAN uses cables, WLAN uses WiFi (wireless)

LAN vs WLAN Comparison:

Feature	LAN	WLAN
Connection Type	Ethernet Cables	Wireless/WiFi
Speed	Very Fast	Fast
Cost	Lower (cables)	Higher (WiFi equipment)
Flexibility	Limited (need cables)	More flexible (anywhere)
Example	Office computers	Home WiFi

4. CAN (Campus Area Network)

What is CAN?

A CAN connects multiple LANs together in a limited area. Think of it as combining several LANs to make a bigger network.

Simple Example: A university with 5 buildings. Each building has its own LAN. All these LANs are connected to make one big network - that's a CAN.

Where CAN is Used:

- Universities
- College campuses
- Large corporate buildings in the same area
- Hospital complexes

Size: Covers several buildings in the same general area - usually up to a few kilometers

5. MAN (Metropolitan Area Network)

What is MAN?

A MAN is bigger than a CAN. It covers a whole city or town. It connects multiple LANs and CANs together over a large area.

Simple Example: All the banks in your city connected together in one network.

How They Connect:

- High-speed cables (like fiber optic)
- These are very fast cables
- Can send data over longer distances

Where MAN is Used:

- City government offices
- Bank branches in a city
- Hospital networks in a city
- Multiple office buildings in a city

Size: Covers a whole city or town - usually 5 to 50 kilometers

6. SAN (Storage Area Network)

What is SAN?

A SAN is a special, fast network just for storing data. It is completely separate from a regular LAN. It stores huge amounts of information in one place.

Simple Example: A bank's special data storage system that keeps all customer information safe and fast.

Parts of a SAN:

- Multiple disk drives (for storage)
- Fast switches
- Powerful servers

Why Use a SAN?

- **Very Fast:** Specialized for data storage only
- **Reliable:** Keeps backups of important data
- **Separate:** Does NOT slow down regular office networks
- **Large:** Can store huge amounts of data

Where SAN is Used:

- Banks and financial companies
- Large hospitals
- Big companies with lots of data

Key Feature: Works alone - not part of the regular LAN

7. WAN (Wide Area Network)

What is WAN?

A WAN is the largest network type. It covers huge distances - countries, continents, or the whole world. It connects many LANs, CANs, and MANs together.

The Best Example: THE INTERNET!

How It Works:

- Uses many types of connections (cables, radio waves, satellites)
- Connects computers all over the world
- You can send email, video calls, download files, etc.

Characteristics:

- **Largest:** Covers entire world
- **Slowest:** Can be slower than LAN because distances are long
- **Expensive:** Costs a lot of money to set up and run
- **Most Important:** Connects people worldwide

Where WAN is Used:

- The Internet (biggest WAN)
- International company networks
- Global bank networks
- Social media companies

Network Sizes Comparison

Network Type	Size Coverage	Main Devices	Speed	Example
PAN	10-30 meters	Phone, Tablet, Watch	Medium	Your Bluetooth headphones
LAN	One Building	Computers, Printers	Very Fast	School computer lab
WLAN	One Building	WiFi devices	Fast	Home WiFi
CAN	Several Buildings	Multiple LANs	Very Fast	University campus
MAN	Whole City	Multiple LANs/CANs	Fast	City bank branches
SAN	Special Data Network	Storage servers	Extremely Fast	Bank data storage
WAN	Worldwide	Everything connected	Medium	The Internet

Network Hierarchy (From Smallest to Largest):

PAN (smallest) → LAN → CAN → MAN → SAN (special) → WAN (largest)

Think of it like this:

PAN = Your pocket

LAN = Your home/office

CAN = Your campus/complex

MAN = Your city

WAN = The entire world

■ IMPORTANT THINGS TO REMEMBER ■

1. Network Types Are Based on SIZE

Networks are different because they cover different areas. Small networks are for one person (PAN). Large networks cover whole cities (MAN) or the world (WAN).

2. PAN is the Smallest

PAN = Personal Area Network. Just your devices (phone, tablet, watch). Uses Bluetooth or USB cables. Only works for 10-30 meters.

3. LAN is for One Building

LAN = Local Area Network. All computers in one school, office, or home. Usually connected with cables through a switch or WiFi router.

4. WLAN is LAN Without Cables

WLAN = Wireless LAN. Same as LAN but uses WiFi instead of cables. This is why your home WiFi is sometimes called WLAN.

5. CAN Connects Multiple LANs

CAN = Campus Area Network. When many buildings (like a university) are connected together. Each building has its own LAN, but they're all connected.

6. MAN Covers a Whole City

MAN = Metropolitan Area Network. Uses fast cables (fiber optic) to connect networks across a city. Much bigger than CAN.

7. SAN is Special for Data Storage

SAN = Storage Area Network. Not like other networks - it's just for storing huge amounts of data. Very fast and separate from regular networks.

8. WAN is the Biggest - It's the Internet

WAN = Wide Area Network. Covers the entire world! The Internet is the biggest WAN. Connects everything together globally.

9. Remember the Order

From smallest to biggest: PAN → LAN → CAN → MAN → WAN

10. The Key Difference

The main thing that changes is: **HOW BIG THE AREA IS**

- Small area (your room) = PAN
- One building = LAN
- Several buildings = CAN
- Whole city = MAN
- Whole world = WAN